

Physics-Constrained Kalman Filtering for Wheel Normal Load Estimation Using Suspension and IMU Measurements

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Abstract—This paper presents a physics-constrained Kalman filtering approach for estimating individual wheel normal loads using suspension and inertial measurement unit (IMU) measurements. The proposed method combines suspension-based wheel-load pseudo-measurements with IMU-based rigid-body equilibrium relations. These relations, including vertical-force, pitch-moment, and roll-moment equilibrium, are incorporated as probabilistic physical constraints, enabling the estimator to retain local sensitivity to suspension-induced load variations while enforcing global physical consistency in the wheel-load distribution. The method is validated through hardware-in-the-loop (HIL) experiments using IPG CarMaker and dSPACE LabBox, scenarios include the ISO double lane change, Nürburgring circuit, and Hockenheim circuit, and is compared with algebraic, quasi-static, suspension-only, and literature-based observer methods. The results demonstrate consistent performance across moderate transient and demanding full-lap conditions, while ablation and uncertainty studies confirm the dominant role of the physics-based constraint update and examine robustness to model and sensor uncertainty.

Index Terms—Wheel normal load estimation, physics-constrained Kalman filtering, vehicle dynamics, suspension modeling, state estimation.

I. INTRODUCTION

ACCURATE estimation of individual wheel normal loads is essential in vehicle dynamics because tire-road friction potential, and thus the achievable longitudinal and lateral tire forces, depends directly on vertical loading. Reliable wheel-load information is therefore important for braking, traction and stability control, rollover prevention, and advanced chassis control systems. This importance is reflected in recent studies on vehicle state and parameter estimation, where accurate force and friction information supports vehicle stability assessment [1], sideslip and state reconstruction [2], and integrated vehicle dynamics control [3]. Related work on tire-road friction estimation further underscores the role of wheel normal load estimation as a prerequisite for reliable tire-force estimation and vehicle control [4], [5]. Direct measurement using wheel force transducers, however, is generally impractical for production vehicles because of cost, durability, and packaging limitations. Consequently, wheel normal loads are typically estimated from standard onboard sensors.

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Observer-based approaches, particularly Kalman-filter-based methods, have been widely studied for production-oriented wheel-load estimation. Early studies showed that body accelerations and suspension measurements can be used to estimate vertical tire forces [6], [7], and later works improved estimation fidelity by considering road grade, suspension sensing, and coupled vehicle-state estimation [8]–[10]. Related work on online grip and tire-force estimation has also emphasized the importance of combining physical modeling with estimator design for real-time vehicle-dynamics functions [11]. Beyond control-oriented applications, accurate wheel normal load estimation can also support ride comfort and safety by improving the rejection of road-induced disturbances and the use of available tire friction [12]. These requirements are challenging because wheel normal loads contain both suspension-driven high-frequency components associated with road excitation and lower-frequency components associated with inertial load transfer during vehicle maneuvers. Practical estimators should therefore retain local sensitivity to suspension-induced load variations while maintaining globally consistent load distribution at the vehicle level.

Existing approaches can be organized using the two requirements that motivate this work: *local suspension sensitivity*, meaning the ability to respond to corner-specific suspension and road excitation, and *four-wheel consistency*, meaning that the four estimated loads jointly satisfy vehicle-level vertical-force, pitch-moment, and roll-moment relations. Rigid-body and algebraic methods lie primarily on the second axis. For example, Alatorre et al. [13] reconstruct the four loads from vehicle-level equilibrium and therefore preserve global load-transfer consistency. Their local sensitivity is limited, however, because wheel-resolved suspension forces do not directly enter the reconstruction. The same global relations also provide only three independent equations for four corner loads, so an additional distribution assumption is required.

Suspension-based observers lie primarily on the first axis. Reduced-order quarter-car and half-car observers [12], [14], [15] use suspension motion to retain sensitivity to local vertical excitation, but their reduced vehicle descriptions do not impose the complete set of four-corner force and moment relations. Similarly, the per-corner stochastic observer of Ding et al. [1] estimates each wheel from its local suspension measurements; because the four corner estimates are formed independently, simultaneous full-vehicle consistency is not inherent in the observer structure.

Multistage methods combine several estimated quantities, but this does not necessarily satisfy both axes in the final wheel-load vector. The vertical-force blocks in [6]–[9] provide inputs to subsequent tire-force or vehicle-state estimation

TABLE I
COMPARISON OF REPRESENTATIVE WHEEL NORMAL-LOAD ESTIMATION APPROACHES.

Approach	Representative studies	Local suspension sensitivity	Four-wheel consistency	Key limitation or feature
Multistage and cascaded observers	[6]–[9], [16], [17]	Method-dependent	Model-mediated	Intermediate estimates couple observer stages, but no explicit simultaneous constraint is applied to the final four-wheel load state.
Rigid-body methods	[13]	Limited	Explicit	Global force and moment balance is retained, but wheel-resolved suspension excitation is not measured directly.
Reduced-order suspension observers	[12], [14], [15]	Yes	No	Quarter- or half-car models do not enforce complete four-corner balance.
Per-corner stochastic observers	[1]	Yes	No	Corners are estimated independently without full-vehicle force and moment consistency.
Data-driven and intelligent-tire methods	[18]–[24]	Method-dependent	Not inherent	Physical consistency is not guaranteed, and additional training data or specialized sensing may be required.
Proposed method	This work	Yes	Explicit, soft	Local suspension information and full-vehicle equilibrium are fused within the same four-wheel state.

stages, whereas Cordeiro et al. [16], [17] explicitly use a delayed interconnected cascade-observer architecture. These studies are therefore grouped as multistage or cascaded methods rather than implying that every cited study uses the same cascade structure. Their four-wheel coupling is model-mediated: consistency arises through the coupled vehicle models and exchanged intermediate estimates, rather than through an explicit simultaneous constraint applied to the final four-wheel load state.

Data-driven, hybrid, and intelligent-tire approaches can capture nonlinear effects that are difficult to model analytically [18]–[24]. Their position on the two axes is architecture-dependent, and four-wheel physical consistency is not guaranteed unless it is introduced explicitly. They may also require specialized tire sensing, adaptation procedures, or extensive training data.

Table I summarizes this two-axis comparison. “Model-mediated” consistency denotes coupling that emerges through a vehicle model or exchange of intermediate observer outputs; it does not mean that the final four-wheel load vector is explicitly corrected to satisfy all three full-vehicle equilibrium relations at each sample. The gap addressed here is therefore a unified state-estimation structure in which the same four-wheel load state receives wheel-resolved suspension pseudo-measurements and probabilistic full-vehicle equilibrium constraints at every sample.

To address these limitations, this paper proposes a physics-constrained Kalman filtering framework for estimating individual wheel normal loads using suspension and inertial measurement unit (IMU) measurements. The proposed method treats the four wheel normal loads as filter states and converts suspension displacement and velocity signals into wheel-load pseudo-measurements using a compact suspension force model with spring, damper, and anti-roll-bar terms, thereby providing local wheel-level sensitivity. IMU-based vertical-force, pitch-moment, and roll-moment equilibrium relations are then used to provide global vehicle-level consistency. Building on constrained Kalman filtering, which provides a principled way to incorporate physical relations into recur-

sive state estimation [25]–[29], these equilibrium relations are introduced as soft probabilistic physical constraints. This formulation allows the estimator to retain local sensitivity to suspension-induced load variations while enforcing global physical consistency in the wheel-load distribution. An aerodynamic vertical-load correction is also included in the equilibrium relations to reduce speed-dependent bias at higher vehicle speeds.

The main contributions of this paper are represented in Fig. 1 and summarized as follows:

- **Physics-constrained Kalman filtering framework:** A two-stage estimator that combines suspension-based wheel-load pseudo-measurements with IMU-based rigid-body equilibrium relations, retaining local suspension sensitivity while enforcing global physical consistency in the wheel-load distribution.
- **Modular suspension-based pseudo-measurement construction:** A compact suspension-side model is used to generate local wheel-level pseudo-measurements from spring, damper, and anti-roll-bar terms. The formulation preserves a modular structure, allowing the suspension model to be refined without altering the Kalman filter state definition or update equations.
- **Aerodynamic vertical-load correction:** Inclusion of aerodynamic vertical loading in the equilibrium relations to reduce speed-dependent estimation bias at higher vehicle speeds.
- **Broad comparative validation:** HIL evaluation against algebraic, quasi-static, suspension-only, and literature-based observer methods in the ISO double lane change, Nürburgring, and Hockenheim scenarios, supplemented by ablation and model/sensor-uncertainty studies.

The remainder of this paper is organized as follows. Section II presents the suspension-based wheel-load model, which is used to construct wheel-load pseudo-measurements. Section III introduces the IMU-based rigid-body equilibrium relations and describes the proposed physics-constrained Kalman filtering framework that combines these relations with the suspension-based pseudo-measurements. Section IV presents

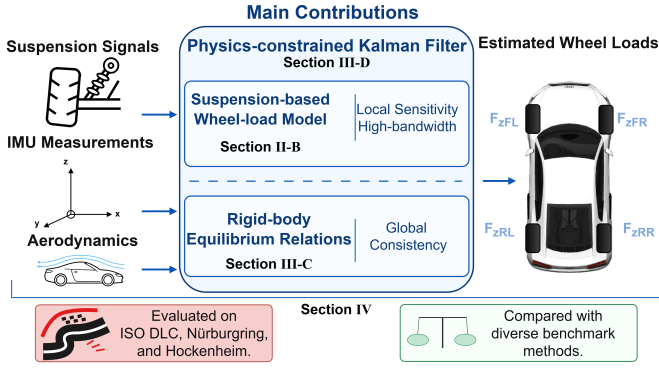


Fig. 1. Main contributions diagram.

the HIL validation, computational assessment, ablation analysis, and robustness studies. Finally, Section V concludes the paper.

II. SUSPENSION-BASED WHEEL-LOAD MODEL

This section develops the suspension-based wheel-load model used in the proposed estimator. The objective is twofold. First, conventional suspension models used in wheel-load estimation are extended to better capture wheel-level load generation mechanisms, including axle coupling and selected asymmetric or nonlinear effects. Second, the resulting suspension-based wheel-load estimates are used as pseudo-measurements in the Kalman filtering framework, providing local wheel-level force information that complements the global rigid-body equilibrium relations introduced in Section III.

Throughout this paper, $F_{z,i}$ denotes the target wheel normal load at corner i within the bandwidth of the estimator. The model developed in this section does not directly reproduce the instantaneous tire-road contact force. Instead, it reconstructs the vertical force transmitted between the suspension and the sprung mass. Explicit tire vertical dynamics are omitted because accurate reconstruction of tire compliance, unsprung-mass acceleration, and wheel-hop motion would require additional tire and wheel measurements that are not normally available from standard onboard sensors.

The suspension-transmitted force nevertheless contains the dominant local information associated with corner-specific compression and rebound. Differences between this force and the tire normal load, principally due to unsprung-mass inertia and tire dynamics, are treated as pseudo-measurement uncertainty rather than assumed to be zero. The Kalman filter therefore does not equate the suspension model output exactly to $F_{z,i}$; R_s determines its statistical weight, while the vehicle-level vertical-force, pitch-moment, and roll-moment relations constrain the final four-wheel state. This fusion allows the suspension model to supply local load variation and the rigid-body relations to correct the total distribution, which explains why accurate wheel-normal-load estimates can be obtained without an explicit tire vertical-dynamics model. Very high-frequency contact-force variations associated with wheel hop remain outside the intended estimation bandwidth.

A. Proposed suspension-based wheel-load modeling

This subsection describes the proposed suspension-based wheel-load model used to construct local wheel-load information for the Kalman filtering framework. Building on conventional quarter-car and half-car suspension representations, the suspension-related contribution to the wheel normal load at each wheel is expressed as

$$F_{z,i}^{\text{susp}} = F_{z,i}^{\text{spr}}(d_i) + F_{z,i}^{\text{dam}}(v_i) + F_{z,i}^{\text{arb}}(\Delta\eta), \quad (1)$$

where $F_{z,i}^{\text{spr}}$, $F_{z,i}^{\text{dam}}$, and $F_{z,i}^{\text{arb}}$ denote the spring, damper, and anti-roll-bar contributions at wheel i , respectively. The variables d_i and v_i denote the measured suspension compression and compression velocity, and $\Delta\eta$ denotes the effective left-right axle deflection driving the anti-roll bar. The wheel index is $i \in \{\text{FL}, \text{FR}, \text{RL}, \text{RR}\}$.

Unlike a classical quarter-car model, this formulation incorporates axle-level coupling through the anti-roll bar. It should therefore be interpreted as an effective wheel-level suspension-load model for constructing local wheel-load information rather than as a full multi-body suspension model. Unsprung-mass dynamics and tire compliance are neglected at this stage. The following subsections define the spring, damper, and anti-roll-bar force contributions used to construct the overall model.

1) *Spring force*: The elastic component of the model is represented by a linear spring law,

$$F_z^{\text{spr}}(d) = k d, \quad (2)$$

where d is suspension compression and k is the axle-specific spring stiffness. This approximation captures the dominant elastic behavior over the normal operating range and keeps the model compact. At larger compression levels, the simulated suspension response may deviate from this model because of bump-stop engagement and other nonlinear effects that are not explicitly represented here.

Separate spring stiffness values are used for the front and rear axles to reflect their different suspension characteristics.

2) *Damper force*: Damper force is modeled as a piecewise-linear function of suspension velocity to account for the asymmetric behavior typically observed between compression and rebound:

$$F_z^{\text{dam}}(v) = \begin{cases} c_{\text{push}} v, & v \geq 0, \\ c_{\text{pull}} v, & v < 0, \end{cases} \quad (3)$$

where $v > 0$ corresponds to compression. The coefficients c_{push} and c_{pull} are obtained from local linear fits to the low-velocity region of the damper force-velocity characteristic.

3) *Anti-roll bar force*: The anti-roll bar (ARB) introduces axle-level coupling between the left and right suspension elements. When the two wheels of an axle experience different vertical displacements, the stabilizer bar generates opposing vertical force contributions at the left and right wheels. This effect provides a physically motivated representation of left-right suspension coupling during lateral maneuvers, where differential suspension motion contributes to wheel-load redistribution. [30].

For a generic axle, the ARB force contribution is modeled in terms of an equivalent differential ARB deformation, $\Delta\eta$, as

$$F_{z,L}^{\text{arb}} = -k_{\text{arb}} \Delta\eta, \quad F_{z,R}^{\text{arb}} = +k_{\text{arb}} \Delta\eta, \quad (4)$$

where k_{arb} denotes the effective ARB stiffness. This sign convention preserves action-reaction across the axle and captures the dominant stabilizer-bar load-transfer effect.

In a real suspension, the equivalent ARB deformation is determined by the left and right suspension motions through the suspension linkage geometry. Therefore, rather than depending on a single suspension displacement, $\Delta\eta$ is generally a non-linear function of both measured suspension compressions,

$$\Delta\eta = \Phi_{\text{arb}}(d_L, d_R), \quad (5)$$

where d_L and d_R are the left and right suspension compressions, and $\Phi_{\text{arb}}(\cdot)$ represents the geometry-dependent ARB deformation mapping.

Because the CarMaker configuration used in this study does not include a detailed ARB linkage-geometry representation, the equivalent differential ARB deformation is modeled using a constant axle-level mapping from the measured suspension compression difference:

$$\Delta\eta_a \approx \gamma_{\text{arb},a} (d_{L(a)} - d_{R(a)}), \quad (6)$$

where $a \in \{f, r\}$ denotes the front or rear axle, and $L(a)$ and $R(a)$ denote the left and right wheels on that axle, respectively.

The dimensionless coefficient $\gamma_{\text{arb},a}$ is an effective motion ratio between the measured suspension-compression difference and equivalent ARB deformation. Because the mapped quantity is a deformation, its exact relation to $d_L - d_R$ can be nonlinear when suspension motion ratios vary with travel. If detailed linkage geometry is available, the mapping can be derived kinematically; alternatively, it can be calibrated by measuring suspension travel vs wheel travel physically. For the operating range considered here, a unity coefficient provides a sufficiently accurate first-order approximation, as supported by the agreement in Fig. 2c. Moreover, the CarMaker ARB model used in this study itself defines the equivalent Hookean stiffness directly with respect to the left-right suspension-compression difference. This simple model relation further supports the adopted values $\gamma_{\text{arb},f} = \gamma_{\text{arb},r} = 1$.

Substituting (6) into (4) yields the ARB force contribution used in the suspension-load model. This term introduces the required left-right axle coupling using the measured suspension compression difference, while any residual discrepancy relative to the CarMaker suspension response is represented through the suspension measurement covariance in the Kalman filter.

Combining the suspension components, the final suspension-based wheel-load estimate at wheel i and discrete time sample k is

$$F_{z,i,k}^{\text{total}} = F_{z,i,0} + F_{z,i}^{\text{spr}}(d_{i,k}) + F_{z,i}^{\text{dam}}(v_{i,k}) + F_{z,i}^{\text{arb}}(\Delta\eta_{a(i),k}), \quad (7)$$

where $d_{i,k}$ and $v_{i,k}$ denote the suspension compression and compression velocity of wheel i at sample k , respectively. The term $a(i)$ denotes the axle associated with wheel i , and

$\Delta\eta_{a(i),k}$ is the corresponding differential ARB deformation input for that axle. The static offset $F_{z,i,0}$ is initialized from the static wheel-load distribution or from the first sample under a nominal operating condition.

The resulting suspension-based wheel-load estimate is used as a pseudo-measurement in the Kalman filtering framework described in Section III. In real suspensions, motion-ratio variation, linkage geometry, bump-stop engagement, and non-linear damper characteristics can produce deviations from the simplified component models, especially over large travel ranges. The present study therefore uses low-order spring, damper, and ARB representations to capture the dominant force trends while maintaining a compact estimator structure. The remaining model mismatch is accounted for through the suspension measurement covariance in the Kalman filtering formulation described in Section III. The spring, damper, and anti-roll-bar force models are illustrated in Fig. 2, together with representative CarMaker-derived force samples.

B. Design rationale

The suspension model is modular with respect to the Kalman filtering framework. Changes to the suspension model affect only the wheel-load pseudo-measurements supplied to the filter, while the state definition, measurement structure, and update equations remain unchanged. The component model is also modular internally, since the spring, damper, and anti-roll-bar effects are constructed as separate additive terms that can be refined independently.

In the present implementation, a linear spring law, a piecewise-linear damper model, and a linear anti-roll-bar term are used to capture the dominant suspension effects with minimal complexity. Among these, the anti-roll-bar term explicitly represents axle-level load coupling, while the asymmetric damper model provides a further refinement of the suspension pseudo-measurements. Their influence on the final estimate depends on the relative weighting of the suspension and physical-constraint updates. If needed, the model can be extended with a full damper map, nonlinear spring behavior, bump-stop terms, or more detailed motion-ratio formulations without changing the Kalman filter state definition or update equations.

III. PHYSICS-CONSTRAINED KALMAN FILTERING FRAMEWORK

A. Estimator overview

The objective of the estimator is to reconstruct the four individual wheel normal loads,

$$\mathbf{x}_k = [F_{z,FL,k} \quad F_{z,FR,k} \quad F_{z,RL,k} \quad F_{z,RR,k}]^T. \quad (8)$$

The state in (8) targets the individual wheel normal loads, whereas the suspension model in Section II-A supplies noisy local proxies derived from suspension-transmitted forces. The second information source is the set of IMU-based rigid-body equilibrium relations, which constrains the total vertical load and its front-rear and left-right distribution. Because both sources are introduced probabilistically, their small physical

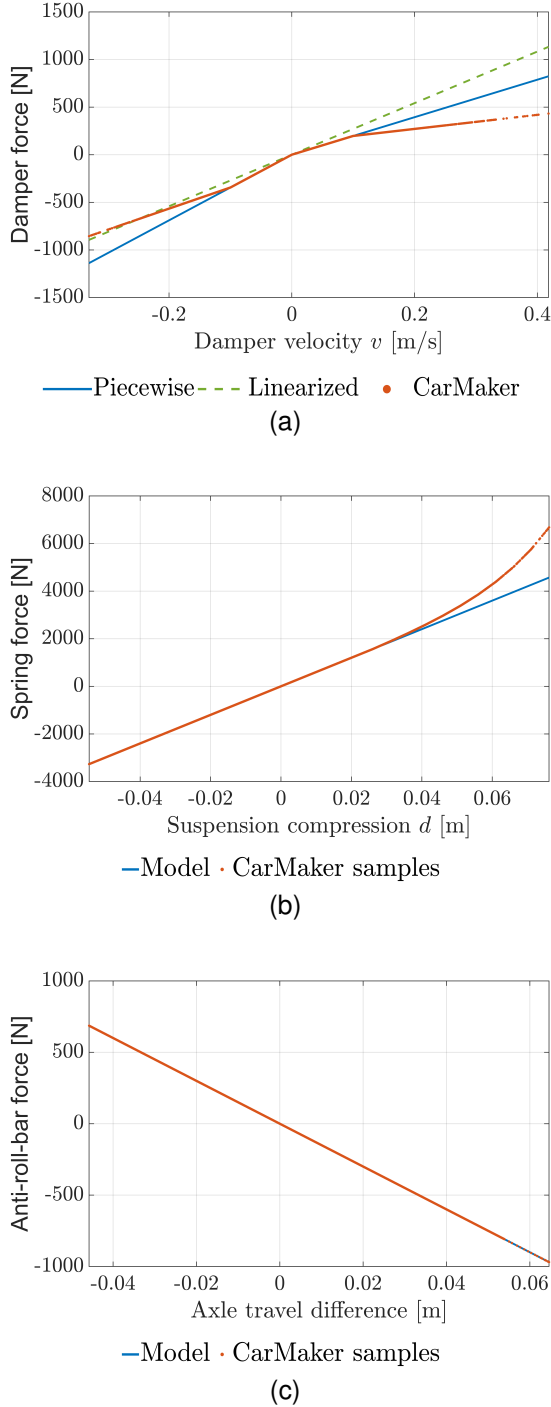


Fig. 2. Proposed suspension models compared with CarMaker-simulated force data: (a) damper force, (b) spring force, and (c) anti-roll-bar force.

mismatch is absorbed through the corresponding covariance models instead of being treated as an exact equality.

These two sources are fused through a sequential two-stage Kalman filtering framework. The suspension-based pseudo-measurements are first used to update the wheel-load states, and the IMU-based equilibrium relations are then imposed as soft probabilistic physical constraints. This structure allows the estimator to retain local sensitivity to suspension-induced load variations while enforcing global physical consistency in the wheel-load distribution.

B. Reference frame and IMU convention

All quantities entering the rigid-body equilibrium relations are expressed in the IMU-aligned body frame $\{\mathcal{B}\}$ to avoid explicit attitude estimation. The axes $(x_{\mathcal{B}}, y_{\mathcal{B}}, z_{\mathcal{B}})$ are aligned with the IMU, with $x_{\mathcal{B}}$ forward, $y_{\mathcal{B}}$ left, and $z_{\mathcal{B}}$ upward.

The accelerometer outputs $(a_{x,k}, a_{y,k}, a_{z,k})$ are interpreted as specific force resolved in $\{\mathcal{B}\}$, meaning that gravity is included. Consequently, for a stationary vehicle on level ground,

$$a_{z,k} \approx g. \quad (9)$$

This convention allows the vertical force balance to be expressed directly using $a_{z,k}$, without reconstructing gravity through explicit attitude estimation. The formulation assumes negligible IMU misalignment or prior alignment compensation.

C. Process and measurement models

The wheel-load state is modeled as a random-walk process,

$$\mathbf{x}_k = \mathbf{x}_{k-1} + \mathbf{w}_{k-1}, \quad \mathbf{w}_{k-1} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}), \quad (10)$$

where $\mathbf{w}_{k-1}$ is the process-noise vector and $\mathbf{Q} \in \mathbb{R}^{4 \times 4}$ is the process-noise covariance matrix.

A zero-order random-walk model was selected because the filter states are the wheel normal loads themselves, and no compact deterministic transition model is available that can accurately propagate the four corner loads under combined road excitation, suspension motion, transient load transfer, aerodynamic loading, and residual modelling errors. The model uses the previous posterior load only as the one-step prior; $\mathbf{Q}$ permits sample-to-sample variation, while the suspension and constraint updates introduce the measured load dynamics. This avoids adding poorly known tire, road, and unsprung-mass states while retaining responsiveness through the selected process covariance [31], [32].

The suspension-based wheel-load estimate developed in Section II-A is used as a pseudo-measurement of the wheel-load state:

$$\begin{aligned} \mathbf{z}_k^{\text{susp}} &= [F_{z,FL,k}^{\text{total}} \quad F_{z,FR,k}^{\text{total}} \quad F_{z,RL,k}^{\text{total}} \quad F_{z,RR,k}^{\text{total}}]^{\top} \\ &= \mathbf{x}_k + \mathbf{v}_{s,k}, \quad \mathbf{v}_{s,k} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_s), \end{aligned} \quad (11)$$

where $\mathbf{z}_k^{\text{susp}}$ denotes the vector of suspension-based wheel-load pseudo-measurements and $\mathbf{R}_s$ is the corresponding covariance. The covariance $\mathbf{R}_s$ represents the uncertainty of the suspension-based pseudo-measurements, including sensor noise and residual mismatch in the suspension-based wheel-load model.

In addition to the local suspension-based pseudo-measurements, the estimated wheel loads are required to satisfy vehicle-level rigid-body equilibrium. Following standard vehicle-dynamics load-transfer formulations [33], [34], the vertical-force, pitch-moment, and roll-moment equilibrium relations can be written compactly as

$$\mathbf{A}\mathbf{x}_k = \mathbf{b}_k, \quad (12)$$

where $\mathbf{A}$ is the rigid-body relation matrix and $\mathbf{b}_k$ is the corresponding equilibrium vector constructed from IMU signals and aerodynamic loading.

The rigid-body relation matrix is

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ l_f & l_f & -l_r & -l_r \\ -\frac{T_f}{2} & \frac{T_f}{2} & -\frac{T_r}{2} & \frac{T_r}{2} \end{bmatrix}, \quad (13)$$

where l_f and l_r are the longitudinal distances from the center of gravity to the front and rear axles, respectively, and T_f and T_r are the front and rear track widths. The corresponding equilibrium vector is

$$\mathbf{b}_k = \begin{bmatrix} ma_{z,k} - F_{z,aero,k} \\ mh_{cg}a_{x,k} + (x_{aero} - x_{cg})F_{z,aero,k} \\ mh_{cg}a_{y,k} \end{bmatrix}, \quad (14)$$

where m is the vehicle mass, h_{cg} is the center-of-gravity height, and x_{cg} and x_{aero} denote the longitudinal positions of the center of gravity and aerodynamic center of pressure, respectively, in the same vehicle-fixed coordinate frame.

Rather than enforcing (12) as an exact equality, the equilibrium relation is introduced as a soft probabilistic physical constraint:

$$\mathbf{b}_k = \mathbf{A}\mathbf{x}_k + \mathbf{w}_{c,k}, \quad \mathbf{w}_{c,k} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_c), \quad (15)$$

where $\mathbf{w}_{c,k}$ represents the constraint uncertainty and $\mathbf{R}_c$ determines the strength of constraint enforcement. This formulation follows the principle of constrained Kalman filtering, in which physical equality relations are incorporated into recursive state estimation while allowing uncertainty in the constraint equations [25], [27], [28].

The aerodynamic vertical force is modeled as

$$F_{z,aero,k} = \frac{1}{2}\rho V_{x,k}^2 A_{ref} c_{L0}, \quad (16)$$

where ρ is the air density, $V_{x,k}$ is the longitudinal vehicle speed, A_{ref} is the aerodynamic reference area, and c_{L0} is the zero-yaw vertical aerodynamic coefficient. Here, $F_{z,aero,k}$ follows the sign convention of (14): positive values correspond to aerodynamic lift and reduce the tire normal-load sum, whereas negative values correspond to aerodynamic downforce. Including this term reduces speed-dependent bias in both the total normal load and its front–rear distribution [33], [35].

D. Two-stage Kalman update

At each time step, the estimator performs one prediction step followed by two sequential measurement updates. The first update uses the suspension-based pseudo-measurements, and the second update applies the IMU-based rigid-body equilibrium relations as soft physical constraints.

1) *Prediction:* From the random-walk process model in (10), the prediction step is

$$\mathbf{x}_{k|k-1} = \mathbf{x}_{k-1|k-1}, \quad \mathbf{P}_{k|k-1} = \mathbf{P}_{k-1|k-1} + \mathbf{Q}, \quad (17)$$

where $\mathbf{P}_{k|k-1}$ is the predicted state-error covariance and $\mathbf{P}_{k-1|k-1}$ is the posterior covariance from the previous time step. The covariance $\mathbf{Q}$ controls the responsiveness of the estimator to wheel-load variations not captured by the random-walk prediction alone.

2) *Suspension-based update:* Based on the suspension pseudo-measurement model in (11), the suspension-update gain is

$$\mathbf{K}_s = \mathbf{P}_{k|k-1} (\mathbf{P}_{k|k-1} + \mathbf{R}_s)^{-1}. \quad (18)$$

The intermediate posterior state estimate is then obtained as

$$\mathbf{x}_k^{(s)} = \mathbf{x}_{k|k-1} + \mathbf{K}_s (\mathbf{z}_k^{\text{susp}} - \mathbf{x}_{k|k-1}), \quad (19)$$

and the corresponding covariance update is

$$\mathbf{P}_k^{(s)} = (\mathbf{I}_4 - \mathbf{K}_s) \mathbf{P}_{k|k-1} (\mathbf{I}_4 - \mathbf{K}_s)^\top + \mathbf{K}_s \mathbf{R}_s \mathbf{K}_s^\top. \quad (20)$$

This first update injects local wheel-resolved force information into the estimator.

3) *Physics-constraint update:* Using the suspension-updated estimate as the prior, the rigid-body equilibrium relations are applied through the soft constraint model in (15). The constraint-update gain is

$$\mathbf{K}_c = \mathbf{P}_k^{(s)} \mathbf{A}^\top (\mathbf{A} \mathbf{P}_k^{(s)} \mathbf{A}^\top + \mathbf{R}_c)^{-1}, \quad (21)$$

and the final posterior state estimate is

$$\mathbf{x}_{k|k} = \mathbf{x}_k^{(s)} + \mathbf{K}_c (\mathbf{b}_k - \mathbf{A} \mathbf{x}_k^{(s)}). \quad (22)$$

The covariance update is

$$\mathbf{P}_{k|k} = (\mathbf{I}_4 - \mathbf{K}_c \mathbf{A}) \mathbf{P}_k^{(s)} (\mathbf{I}_4 - \mathbf{K}_c \mathbf{A})^\top + \mathbf{K}_c \mathbf{R}_c \mathbf{K}_c^\top. \quad (23)$$

This second update regularizes the wheel-load estimates by enforcing global vehicle-level force and moment consistency.

Because the suspension pseudo-measurement model and the rigid-body constraint model are both linear, and their measurement errors are assumed to be mutually independent, the sequential and jointly stacked Kalman updates are algebraically equivalent and yield the same posterior state estimate and covariance. Therefore, the sequential implementation introduces no theoretical loss of optimality and is not intended to provide an inherent estimation-performance advantage over the corresponding joint update. The sequential form is retained primarily for modular implementation and methodological clarity, as it explicitly separates the incorporation of local wheel-resolved suspension information from the subsequent enforcement of global force- and moment-equilibrium consistency.

E. Measurement structure and interpretation

The suspension pseudo-measurement model in (11) provides one direct pseudo-measurement for each wheel normal load. Therefore, the suspension update supplies wheel-resolved information for all four states. In contrast, the rigid-body constraint model in (15) provides three vehicle-level consistency relations: vertical-force, pitch-moment, and roll-moment equilibrium.

The constraint matrix $\mathbf{A} \in \mathbb{R}^{3 \times 4}$ has rank three under normal vehicle geometry. Thus, the rigid-body equilibrium relations alone cannot uniquely determine the four wheel normal loads. Instead, they provide three independent physical consistency conditions that couple the corner loads through global vehicle dynamics.

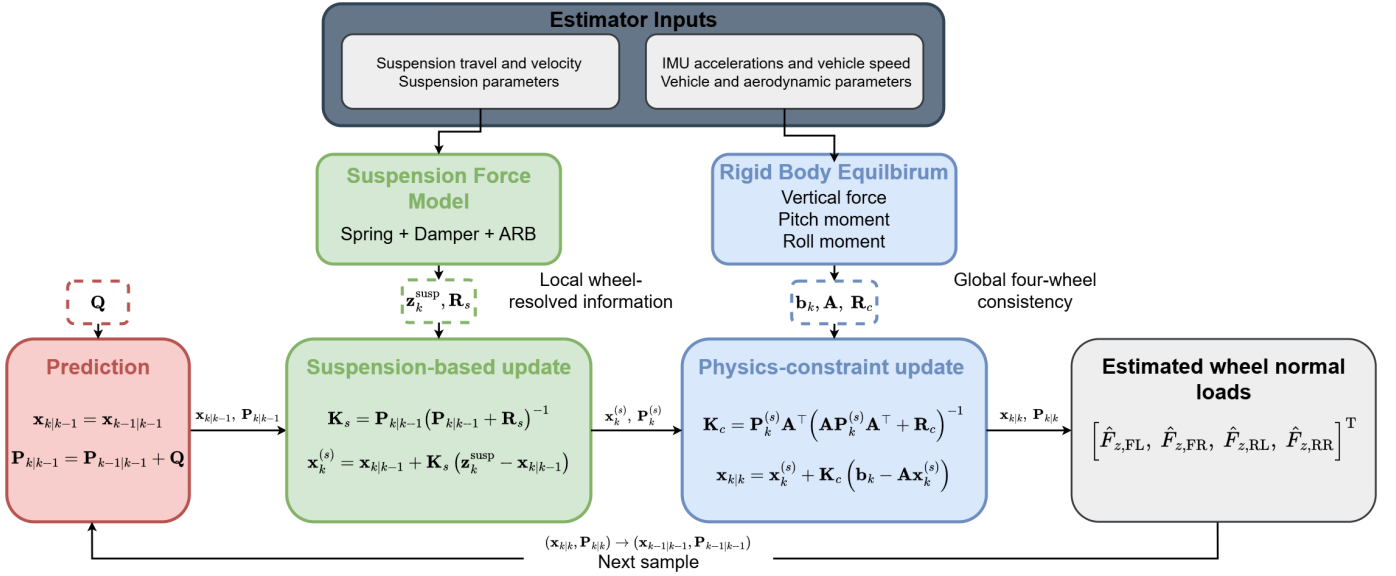


Fig. 3. Block-diagram representation of the proposed wheel-load estimator. Suspension measurements are first converted into wheel-level pseudo-measurements, which are fused through a prediction step and a suspension update. IMU-based vertical-force, pitch-moment, and roll-moment equilibrium relations are then applied as soft constraints in a second update stage.

Consequently, the primary role of the constraint update is not to recover the wheel normal loads by itself, but to regularize the suspension-based local estimates by enforcing physically consistent load redistribution across the vehicle. This interpretation is central to the proposed framework: the suspension update provides local wheel-level sensitivity, while the constraint update provides global physical consistency.

F. Implementation details

The estimator is implemented as a sequential two-stage Kalman filter. The per-sample estimation cycle is summarized in Fig. 3.

G. Covariance tuning procedure

1) *Tuning procedure used in this study:* The covariance parameters were investigated through two consecutive sweeps using the Nürburgring scenario. The first examined the common process-noise variance Q applied to all four wheel-force states. Its corresponding standard deviation, $\sqrt{Q}$, was varied over the range shown in Fig. 4a.

After fixing Q , the relative weighting of the suspension pseudo-measurements and rigid-body constraints was investigated. The constraint standard deviations were expressed as a common percentage of the respective nominal vertical-force, pitch-moment, and roll-moment magnitudes. This preserves their physically determined relationship while accounting collectively for IMU uncertainty, vehicle-parameter uncertainty, the rigid-body approximation, and aerodynamic-model error.

The same suspension standard deviation was used for all four wheels and expressed as a percentage of a representative nominal wheel load. This avoids wheel-dependent tuning parameters and represents the combined effects of displacement-measurement error, numerical differentiation, and suspension-model mismatch. The suspension and constraint percentages

were then varied simultaneously to generate Fig. 4b. The selected operating point corresponds to suspension and constraint uncertainties of approximately 40% and 10%, respectively. The resulting covariance matrices were subsequently held fixed for every validation scenario.

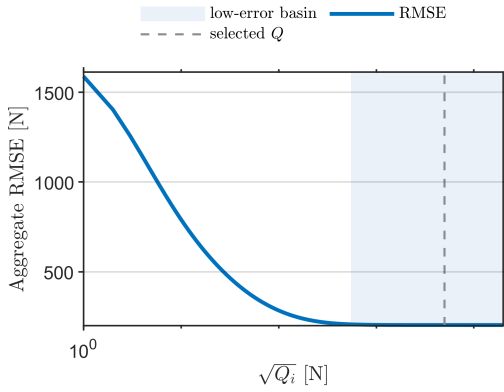
2) *Sensitivity interpretation and recommended baseline:* Figure 4a shows that $\sqrt{Q}$ must be sufficiently large for the estimator to respond to changes indicated by the suspension measurements and rigid-body constraints. At smaller values, the previous wheel-force estimate remains overly influential. Above this threshold, the error enters a broad plateau because the two measurement updates dominate the corrected estimate. A value within this plateau was therefore selected.

The two-dimensional sweep in Fig. 4b exhibits an approximately linear low-error valley, indicating that performance depends primarily on the relative weighting of the two updates. The valley widens as both uncertainty magnitudes increase, reducing sensitivity to small tuning changes. The physically initialized constraint uncertainty of 10% intersects this region at a suspension uncertainty of approximately 40%. This point provides a sufficiently wide low-error region without requiring uncertainty magnitudes that are difficult to interpret physically. These percentages may therefore provide a practical initial tuning baseline for implementations with comparable measurements and model structure.

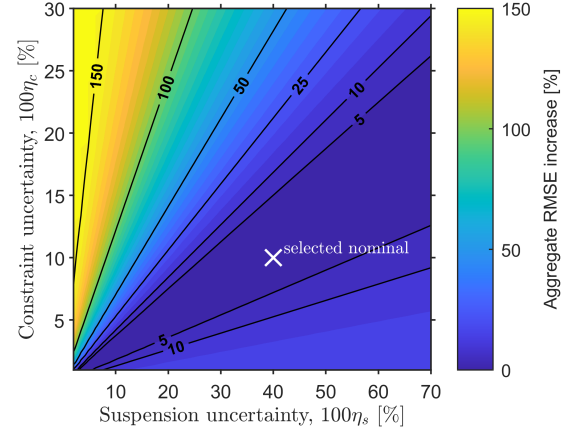
The final standard-deviation vectors were

$$\begin{aligned}\sigma_Q &= [5.0, 5.0, 5.0, 5.0]^\top \times 10^3 \text{ N}, \\ \sigma_s &= [2.0, 2.0, 2.0, 2.0]^\top \times 10^3 \text{ N}, \\ \sigma_c &= [1600.0 \text{ N}, 3300.0 \text{ N m}, 1000.0 \text{ N m}]^\top.\end{aligned}\quad (24)$$

The diagonal covariance matrices Q , R_s , and R_c were formed by element-wise squaring of the corresponding standard-deviation vectors.



(a) Process-noise sweep.



(b) Suspension and constraint weighting sweep.

Fig. 4. Covariance sensitivity for the Nürburgring tuning case: (a) aggregate RMSE versus the common process-noise standard deviation $\sqrt{Q}$; and (b) aggregate RMSE increase versus the suspension and constraint standard deviations expressed as percentages of their respective nominal magnitudes. The marker denotes the selected 40% suspension and 10% constraint uncertainties.

IV. HARDWARE-IN-THE-LOOP VALIDATION AND ROBUSTNESS ANALYSIS

A. HIL environment and test vehicle

The comparative validation was reproduced using a real-time signal-replay HIL configuration comprising IPG CarMaker 15 (CarMaker for Simulink 15.0) and a dSPACE LabBox. CarMaker provided the virtual vehicle plant, sensor channels, and independent reference wheel loads, while the proposed estimator and all five comparison methods were compiled and executed on the LabBox. The vehicle was based on the built-in Audi R8 2016 model with a 4WD driveline.

The ISO double lane change, Nürburgring, and Hockenheim signal sequences were replayed to the hardware at a fixed sampling period of $\Delta t = 0.0\text{ s}$, corresponding to a 100.0 Hz estimator task and a 10.0 ms deadline. Each estimator executed once per input sample using the subset of acceleration, suspension displacement, wheel translation, and vehicle-speed signals required by its formulation. The CarMaker wheel-load channels were not supplied to any estimator; they were retained only for post-test error calculation. This arrangement evaluates the numerical estimators on physical real-time hardware under repeatable inputs, while retaining controlled access to a common four-wheel reference.

Key geometric, inertial, aerodynamic, and suspension parameters extracted from the vehicle information file are summarized in Table II. These parameters were used both to configure the proposed estimator and to define the baseline methods in a consistent manner, ensuring a fair comparison across all approaches.

B. Evaluation setup, compared methods, and metrics

The validation is structured to assess the proposed estimator from complementary perspectives. The ISO double lane change represents a moderate transient handling condition with pronounced but short-duration lateral suspension excitation. The Nürburgring and Hockenheim full-lap scenarios assess robustness under sustained combined maneuvers and elevation

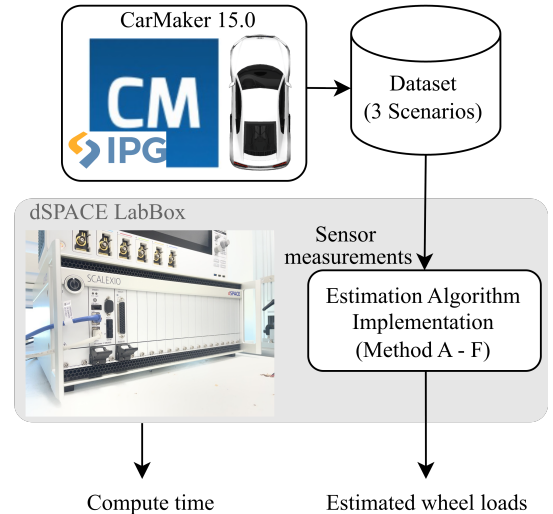


Fig. 5. dSPACE LabBox HIL experimental setup used for real-time estimator execution.

changes, with the Nürburgring providing the stronger coupled excitation. Together, these HIL experiments examine whether the estimator can retain local sensitivity to suspension-induced load variations while maintaining global physical consistency in the wheel-load distribution.

All estimators were executed on the LabBox using identical replayed CarMaker input sequences and the four CarMaker wheel normal loads $\{F_{z,FL}, F_{z,FR}, F_{z,RL}, F_{z,RR}\}$ as reference. To ensure fairness, all baseline methods were implemented using the same vehicle parameters and sensor signals. Where necessary, parameters were tuned within physically reasonable ranges to provide each method with comparable operating conditions.

The following methods were compared:

- **Method A:** the **proposed** physics-constrained Kalman filter, combining suspension-based wheel-load pseudo-

TABLE II
PRIMARY CARMAKER VEHICLE PARAMETERS USED IN THE HIL VALIDATION.

Parameter	Symbol	Value
Total vehicle mass	m	1644.8 kg
CG / body reference location	(x_{cg}, y_{cg}, z_{cg})	(2.267, 0, 0.497) m
Body inertia about reference	(I_{xx}, I_{yy}, I_{zz})	(518.04, 1677.0, 1921.3) kg m ²
Wheelbase	L	2.750 m
CG distances to axles	(l_f, l_r)	(1.223, 1.527) m
Track widths	(T_f, T_r)	(1.632, 1.582) m
Tires	–	Front 245/35 R19; rear 295/35 R19
Aerodynamic reference area	A_{ref}	2.1 m ²
Aerodynamic application point	(x_{aero}, y, z)	(2.08, 0, 0.62) m
Nominal lift coefficient	c_{L0}	0.149 at 0° yaw
Front spring rate	k_f	60 000.0 N/m
Rear spring rate	k_r	70 000.0 N/m
Anti-roll bar stiffness	$(k_{arb,f}, k_{arb,r})$	15 000.0 N/m front and rear, Hookean
Front damper force map	$F_{dam,f}(v)$	Push: (0.1 m/s, 196.9 N), (1.0 m/s, 861.97 N); pull: (0.1 m/s, 344.69 N), (1.0 m/s, 2339.9 N)
Rear damper force map	$F_{dam,r}(v)$	Push: (0.1 m/s, 198.66 N), (1.0 m/s, 879.62 N); pull: (0.1 m/s, 312.15 N), (1.0 m/s, 2014.6 N)
Bump-stop engagement	$(d_{0,f}, d_{0,r})$	(0.026, 0.040) m, buffer $\tau \geq 0$
IMU sensor location	(x, y, z)	(2.107, 0, 0.497) m

measurements with IMU-based rigid-body equilibrium constraints;

- **Method B:** an algebraic normal-load estimator based on the formulation of Alatorre et al. [13], in which road grade and bank are approximated from low-frequency suspension-derived pitch and roll;
- **Method C:** a rigid-body quasi-static (QS) load-transfer baseline anchored to static axle loads;
- **Method D:** a suspension-only baseline corresponding to the proposed suspension-based wheel-load model evaluated without Kalman filtering or explicit constraint enforcement;
- **Method E:** a quarter-car vertical tire-force observer based on Cheon et al. [12], adapted to the available full-car CarMaker signals by reconstructing per-corner inputs from the body IMU and suspension measurements;
- **Method F:** a per-corner vertical tire-force estimator based on Ding et al. [1], implemented as a strong-tracking unscented Kalman filter (STUKF) using suspension compression and compression-rate measurements, and adapted to the available CarMaker suspension signals.

These methods were selected to test the proposed estimator against complementary baseline classes that reflect the main trade-offs in wheel normal load estimation. Method B represents a rigid-body-equilibrium-based estimator and was included because this class directly addresses vehicle-level force and moment balance. Method C provides a quasi-static load-transfer baseline, which remains a standard reference in many practical vehicle-dynamics frameworks due to its simplicity and low computational cost. Method D represents a suspension-only formulation and was included to isolate the contribution of the suspension-based wheel-load model independently of Kalman filtering and explicit constraint enforcement. Methods E and F were selected as recent literature-based observer formulations. Method E corresponds to a reduced-order suspension-based observer, whereas Method F represents a stochastic filtering approach based on suspension

measurements. Collectively, these baselines span rigid-body, quasi-static, suspension-only, reduced-order observer-based, and stochastic filtering approaches, thereby providing a diverse comparison across methods that emphasize local suspension information, global load-transfer relations, or observer-based fusion.

The direct numerical comparison was restricted to methods that could be reproduced using the common suspension and inertial signals available in the CarMaker dataset. More elaborate interconnected cascade estimators, such as Cordeiro et al. [16], [17], were included in the broader comparison in Table I but were not implemented numerically.

A faithful reconstruction of such methods requires multiple coupled observers, delayed exchange of intermediate longitudinal, lateral, and vertical force estimates, and consistent specification of the complete vehicle model, observer states, initialization, covariance matrices, and stage-specific tuning. An approximate implementation would introduce substantial reconstruction-dependent uncertainty and would not provide a controlled comparison with the published method. Similarly, data-driven and intelligent-tire methods were not implemented because they require method-specific training datasets or specialized tire measurements unavailable in the common dataset.

Validation is presented in two stages of increasing complexity. First, an ISO double lane change is used as a standardized transient-handling maneuver with pronounced lateral load transfer. Second, full-lap results on the Nürburgring and Hockenheim circuits evaluate estimator performance under sustained combined excitation, elevation change, and track-dependent maneuver severity.

Let the wheel-load estimation error for wheel $i \in \{\text{FL}, \text{FR}, \text{RL}, \text{RR}\}$ be defined as

$$e_i(t_k) = \hat{F}_{z,i}(t_k) - F_{z,i}^{\text{ref}}(t_k). \quad (25)$$

Performance is primarily reported using per-wheel error metrics over a common evaluation window. For each wheel,

the root-mean-square error (RMSE), 95th-percentile absolute error, and maximum absolute error are defined as

$$\text{RMSE}_i = \sqrt{\frac{1}{N} \sum_{k=1}^N e_i^2(t_k)}, \quad (26)$$

$$P95_i = \text{percentile}_{95}(|e_i(t_k)|), \quad (27)$$

$$\text{Max}_i = \max_k |e_i(t_k)|, \quad (28)$$

where N is the number of samples in the evaluation window.

In addition, a normalized root-mean-square error is reported as

$$\text{RMSE}_{\%,i} = 100 \times \frac{\text{RMSE}_i}{\overline{F_{z,i}^{\text{ref}}}}, \quad (29)$$

where $\overline{F_{z,i}^{\text{ref}}}$ denotes the mean reference wheel normal load for wheel i over the same evaluation window.

For the ablation and Monte Carlo studies, aggregate metrics are also reported by pooling errors across all four wheels and all samples. These pooled metrics are used to summarize overall estimator robustness without emphasizing wheel-specific effects. The aggregate RMSE, 95th-percentile absolute error, and maximum absolute error are defined as

$$\text{RMSE}_{\text{agg}} = \sqrt{\frac{1}{4N} \sum_{i=1}^4 \sum_{k=1}^N e_i^2(t_k)}, \quad (30)$$

$$P95_{\text{agg}} = \text{percentile}_{95}(\{|e_i(t_k)|\}_{i=1,\dots,4; k=1,\dots,N}), \quad (31)$$

$$\text{Max}_{\text{agg}} = \max_{i,k} |e_i(t_k)|. \quad (32)$$

When reported, the aggregate percentage RMSE is defined as

$$\text{RMSE}_{\%,\text{agg}} = 100 \times \frac{\text{RMSE}_{\text{agg}}}{\frac{1}{4} \sum_{i=1}^4 \overline{F_{z,i}^{\text{ref}}}}. \quad (33)$$

For the Monte Carlo noise study, these aggregate metrics are evaluated for each trial and then summarized across trials using the median and 95th percentile.

C. ISO double lane change HIL evaluation

Table III summarizes the HIL wheel-load estimation errors over the trimmed maneuver interval (5.0 s to 12.0 s), with corresponding time-history comparisons shown in Fig. 6. Method D achieves the lowest errors on all four wheels, followed by Method F, while Method A provides the strongest performance among Methods A–C and E. The pooled RMSE values are 51.3 N, 59.4 N, and 101.8 N for Methods D, F, and A, respectively.

The ISO DLC considered here should be interpreted as a moderate transient condition rather than a severe combined-excitation case. Methods A and D use the same suspension pseudo-measurements, but Method A subsequently applies an instantaneous correction based on the three rigid-body equilibrium relations. During the short DLC interval, the local suspension model already follows the dominant corner-load variation closely, whereas the suspension peaks and filtered

TABLE III
WHEEL-LOAD ESTIMATION ERROR METRICS OVER THE DLC HIL
SCENARIO FROM 5.0 TO 12.0 s.

Metric	Method	FL	FR	RL	RR
RMSE [N]	A	111.53	112.27	88.82	92.17
	B	164.96	164.89	108.60	111.33
	C	150.46	150.09	135.95	138.02
	D	56.77	56.44	45.41	45.50
	E	239.37	239.09	211.93	210.94
	F	63.22	63.44	54.21	55.94
RMSE [%]	A	2.72	2.74	2.26	2.34
	B	4.03	4.02	2.76	2.83
	C	3.67	3.66	3.45	3.50
	D	1.39	1.38	1.15	1.15
	E	5.84	5.83	5.38	5.35
	F	1.54	1.55	1.38	1.42
P95 [N]	A	214.75	231.77	161.58	161.78
	B	313.47	326.14	216.72	215.65
	C	300.45	299.94	260.65	272.84
	D	115.86	118.09	97.82	99.19
	E	372.13	375.12	341.35	342.71
	F	133.51	139.01	118.69	128.86
MAX [N]	A	391.51	411.17	367.00	371.05
	B	563.94	581.27	466.61	439.18
	C	414.09	409.27	601.45	575.56
	D	163.01	158.15	161.52	162.58
	E	567.12	567.33	856.39	854.18
	F	197.34	210.00	216.06	200.16

body-IMU equilibrium quantities do not coincide exactly. The constraint update therefore trades some maneuver-specific peak accuracy for simultaneous four-wheel consistency.

Under stronger and more sustained combined transients, local suspension-model mismatch and cross-vehicle load redistribution become more influential. The equilibrium residual then contains more useful corrective information, so the physics-constraint update becomes more effective, as reflected by the Nürburgring HIL results in Section IV-D. This explains why the proposed method need not be best in the moderate DLC to provide the strongest performance under the more demanding full-lap condition.

Overall, the ISO double lane change serves as a moderate, isolated transient benchmark. While Methods D and F provide the lowest maneuver-specific errors, Method A retains the intended trade-off between local peak following and physically consistent four-wheel load distribution.

D. Full-lap HIL evaluation

Following the controlled ISO DLC, estimator performance is evaluated over complete laps of the Nürburgring Nordschleife and Hockenheim circuits.

These scenarios are included to assess robustness under extended high-excitation driving conditions that are not fully represented by a single standardized maneuver. The Nürburgring Nordschleife is widely recognized as an industrial vehicle-development and validation environment, with manufacturers

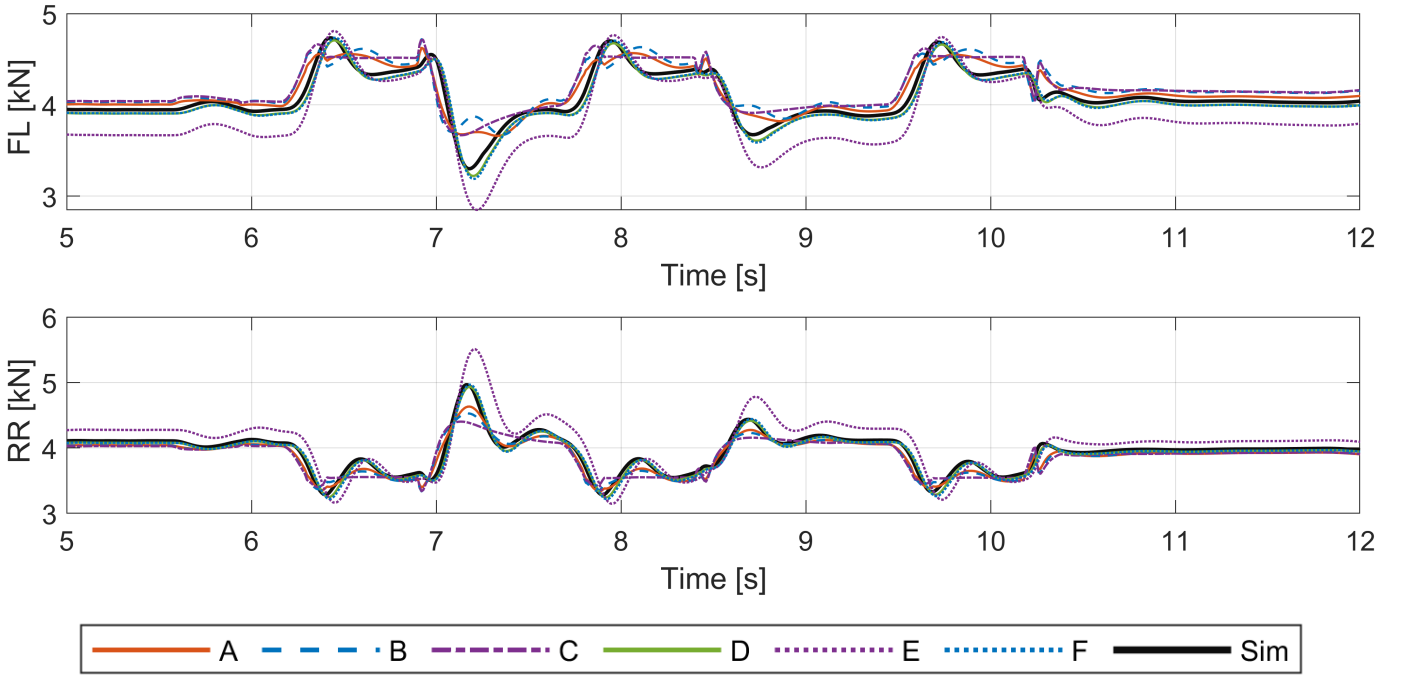


Fig. 6. HIL-estimated and CarMaker-reference wheel normal loads during the trimmed ISO double lane change maneuver.

and suppliers using the circuit to evaluate vehicle, component, chassis, aerodynamic, and control-system behaviour under severe operating conditions [36]. In addition, prior vehicle-dynamics studies have used the Nürburgring for chassis-control development, high-performance vehicle simulation, and full-lap dynamic analysis on complex three-dimensional track geometries [37], [38].

The Nürburgring case is therefore used here as the more demanding validation scenario because the lap contains sustained combined longitudinal and lateral acceleration, significant elevation variation, changing curvature, and road-bank/surface variations. These features produce repeated load-transfer transients and coupled pitch-roll excitation, which directly challenge wheel-load estimation accuracy. Hockenheim is used as a complementary circuit case, providing a smoother and more uniform excitation profile while still including high-speed braking, acceleration, and cornering events.

Representative track sections used in the evaluation are shown in Fig. 7. The Nürburgring segments in Fig. 7(a)–(c) illustrate the more severe combined-maneuver and elevation-changing conditions, whereas Hockenheim provides a less irregular but still dynamically relevant comparison case, as shown in Fig. 7(d).

Tables IV and V summarize the full-lap HIL error statistics.

On the Nürburgring, Method A achieves the lowest RMSE and P_{95} error on all four wheels. Its RMSE ranges from 209.7 N to 283.4 N, corresponding to 4.9% to 7.2% of the mean corner loads, and its pooled RMSE is 234.6 N. Method B is the next strongest method, with a pooled RMSE of 375.3 N. Methods D and F, which performed strongly in the moderate ISO DLC, deteriorate to RMSE values of 524.9 N and 822.2 N, respectively. Method D records the lowest isolated maximum errors at the two rear wheels, but Method A has substantially

lower RMSE and P_{95} errors, indicating more reliable performance over the complete lap.

On Hockenheim, all methods exhibit lower errors owing to the smoother excitation profile. Method A again provides the lowest RMSE and P_{95} error on every wheel, with RMSE values from 72.9 N to 100.8 N, or approximately 1.6% to 2.9% of the mean corner loads. Its pooled RMSE is 83.7 N, compared with 134.3 N for Method B. Methods C–F produce larger errors across the lap, confirming that the advantage of the constraint update is retained under a less irregular but still dynamically relevant circuit condition.

Figure 8 illustrates representative front-left wheel normal-load histories recorded from the HIL tests for both circuits. Method A closely tracks the CarMaker reference, Method B captures the general trend but exhibits larger transient deviations, and the reduced-order and per-corner methods remain more sensitive to sustained model mismatch.

Overall, the full-lap HIL results confirm that suspension-dominated approaches can perform well in a moderate isolated maneuver, whereas the proposed physics-constrained framework becomes more effective under sustained coupled excitation involving global load redistribution.

E. Real-time implementation feasibility

Execution times were measured under the same generated-code and real-time task configuration used for the HIL evaluation. Each estimator was profiled over more than 50,000 executions of the Nürburgring sequence at 100.0 Hz. As shown in Table VI, the proposed estimator required a mean execution time of 2.7 μ s, a 99th-percentile time of 3.0 μ s, and a maximum of 3.5 μ s. Its mean execution time was approximately 40.0% higher than that of Method B, 66.0% higher than those of Methods C and D, and 28.0% lower than that of Method F.

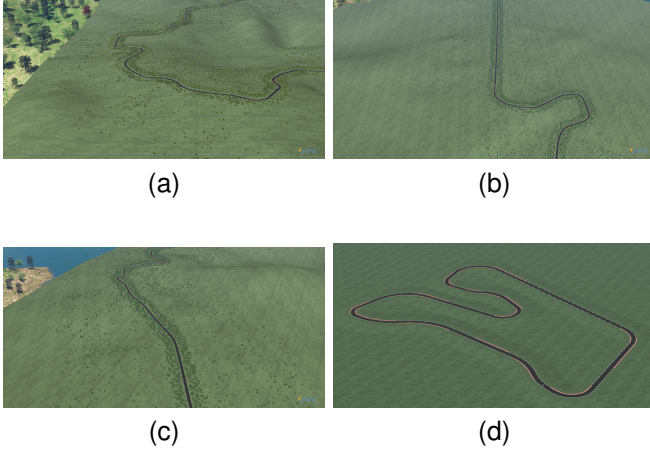


Fig. 7. Representative images of the evaluated track sections. (a) Segment 09: Karussell. (b) Segment 11: Kleine Karussell. (c) Segment 12: Antoniusbuche. (d) Hockenheim circuit.

TABLE IV
WHEEL-LOAD ESTIMATION ERROR METRICS OVER THE NÜRBURGRING HIL SCENARIO.

Metric	Method	FL	FR	RL	RR
RMSE [N]	A	220.46	217.30	283.40	209.72
	B	496.74	328.75	388.70	239.87
	C	800.77	774.17	671.12	650.83
	D	573.99	617.06	459.59	424.95
	E	728.93	683.98	453.75	441.85
	F	890.33	908.42	755.88	717.53
RMSE [%]	A	4.94	5.11	7.21	5.66
	B	11.12	7.73	9.90	6.47
	C	17.93	18.21	17.08	17.57
	D	12.85	14.51	11.70	11.47
	E	16.32	16.09	11.55	11.93
	F	19.93	21.37	19.24	19.37
P95 [N]	A	442.88	430.22	531.93	412.09
	B	885.94	665.72	739.53	475.63
	C	1666.33	1569.46	1369.77	1355.28
	D	1087.87	1240.66	904.96	872.36
	E	1305.67	1310.89	894.35	880.99
	F	1597.97	1729.13	1418.74	1282.90
MAX [N]	A	2174.97	2154.05	1747.56	2053.84
	B	2363.53	2393.59	1853.21	1616.89
	C	5368.72	5150.58	5511.97	4846.84
	D	2175.96	3386.80	1601.06	1547.77
	E	2114.91	2329.37	2547.14	2112.02
	F	2650.68	3880.66	2011.04	2207.49

The fixed four-state structure also gives bounded memory and arithmetic requirements. These results demonstrate real-time feasibility and a low comparative computational burden on the LabBox. However, because the LabBox is substantially more powerful than a production automotive microcontroller, the absolute timings should not be interpreted as production-ECU benchmarks.

TABLE V
WHEEL-LOAD ESTIMATION ERROR METRICS OVER THE HOCKENHEIM HIL SCENARIO.

Metric	Method	FL	FR	RL	RR
RMSE [N]	A	74.37	72.85	83.81	100.80
	B	156.56	146.39	102.38	125.34
	C	194.22	186.71	170.32	141.63
	D	276.83	215.95	190.51	169.42
	E	336.40	264.83	187.17	163.26
	F	427.50	355.49	337.66	310.61
RMSE [%]	A	1.61	1.75	2.15	2.91
	B	3.40	3.51	2.63	3.62
	C	4.22	4.48	4.37	4.09
	D	6.01	5.18	4.89	4.89
	E	7.30	6.35	4.80	4.71
	F	9.28	8.52	8.66	8.96
P95 [N]	A	146.64	112.32	146.67	151.29
	B	245.78	241.20	196.83	198.57
	C	296.91	286.89	282.96	295.03
	D	452.77	416.26	312.41	318.09
	E	483.34	471.57	312.32	259.07
	F	638.27	590.46	524.64	492.70
MAX [N]	A	709.40	733.65	1004.77	674.62
	B	780.72	1394.70	974.85	727.58
	C	1137.03	2174.10	1623.30	1112.87
	D	837.14	754.74	658.49	690.59
	E	1025.84	910.02	757.25	736.31
	F	1017.74	947.48	724.22	844.75

TABLE VI
EXECUTION-TIME STATISTICS MEASURED ON THE DSPACE LABBOX AT 100.0 Hz.

Estimator	Mean [μ s]	P_{99} [μ s]	Max. [μ s]	Relative mean
A	2.69	3.02	3.48	1.00
B	1.92	2.25	2.63	0.71
C	1.62	1.96	2.36	0.60
D	1.62	1.95	2.14	0.60
E	1.67	1.99	2.37	0.62
F	3.73	4.09	4.40	1.39

F. Ablation analysis

Controlled MATLAB ablations were performed using the same Nürburgring input, 100.0 Hz sampling rate, causal pre-processing, initialization, and covariance settings as the HIL implementation but conducted offline inside of simulink. Only the component specified in each study was changed.

Table VII summarizes the contribution of each estimator component at the selected covariance setting. Removing aerodynamic compensation increased the aggregate RMSE from 233.0 to 237.9 N (2.1 %) and the P_{95} error from 460.3 to 496.2 N (7.8 %). The comparatively larger change in P_{95} is consistent with aerodynamic compensation reducing some upper-tail, speed-dependent errors, although its effect on the aggregate RMSE was modest. Disabling the anti-roll-bar model increased the RMSE to 236.4 N (1.5 %) and the P_{95} error to 465.9 N, showing a similarly modest incremental benefit at the selected covariance setting. These results assess

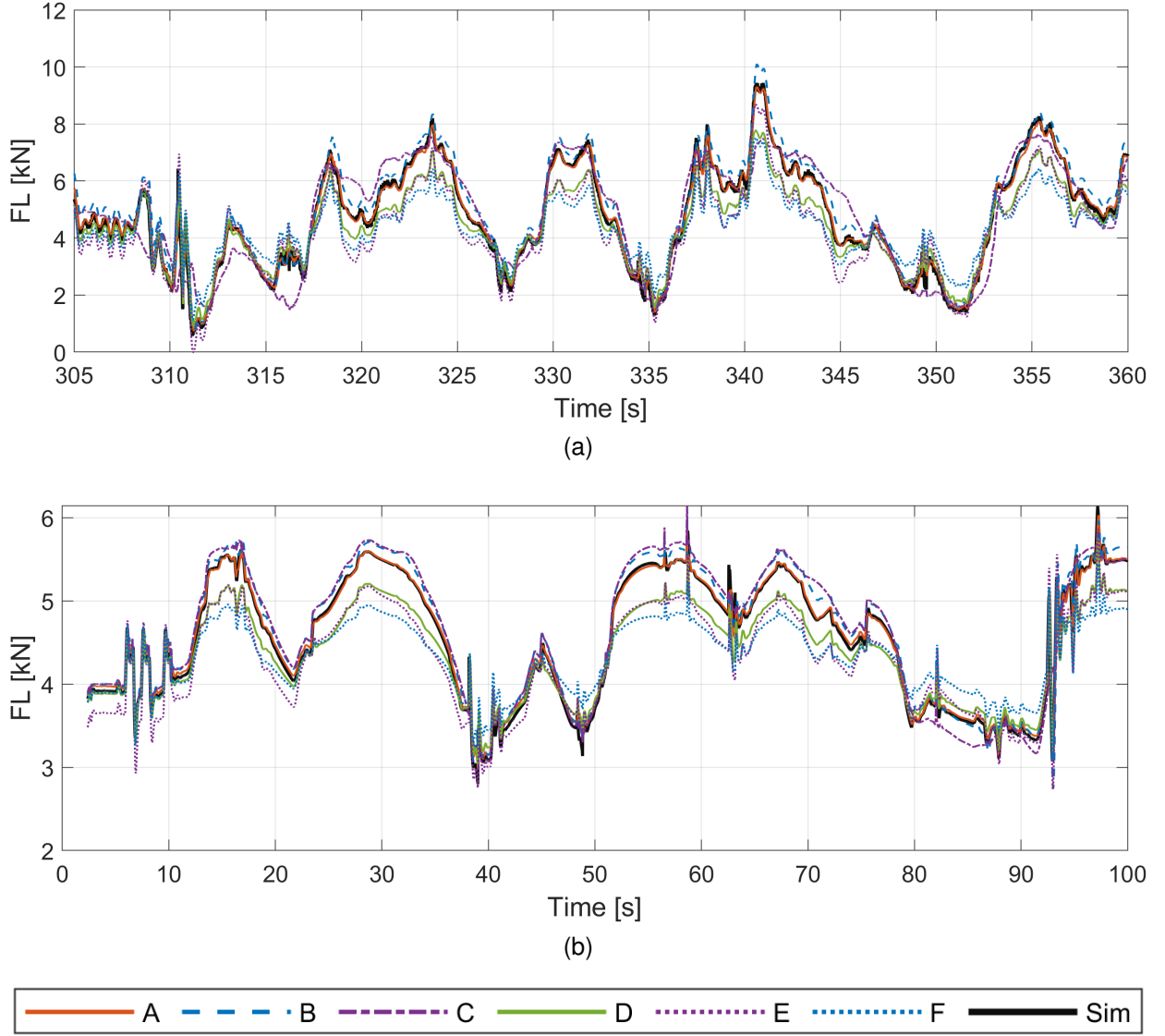


Fig. 8. Comparison of wheel normal-load estimation methods for the selected driving cases. (a) Karussell segment from the Nürburgring HIL case. (b) Single lap from the Hockenheim HIL case.

each component only at the selected covariance setting. The aerodynamic and anti-roll-bar terms may additionally reduce structured model mismatch and thereby broaden the range of effective weightings between R_s and R_c ; however, establishing this effect would require component-specific covariance sweeps and is not claimed from the present ablation results.

Replacing the asymmetric damper model with a symmetric linear model had negligible influence: the aggregate RMSE changed from 233.0 to 232.9 N and the P_{95} error from 460.3 to 460.0 N. Damper asymmetry is therefore retained as a physical refinement but was not a material contributor in this scenario.

Removing the rigid-body constraints caused the largest degradation, increasing the aggregate RMSE from 233.0 to 523.0 N (124.5%), the P_{95} error from 460.3 to 1053.3 N (128.8%), and the maximum error from 2161.2 to 3324.3 N. This confirms that the constraint update is essential for maintaining global four-wheel force and moment consistency.

G. Robustness to model and sensor uncertainty

Model-parameter and sensor uncertainties representative of real-vehicle operation were introduced using the Nürburgring input. The causal preprocessing, initialization, sampling rate, and Kalman-filter covariance settings used in the HIL implementation were retained so that only the specified uncertainty source changed.

a) *Model-parameter sensitivity*: Each parameter was perturbed independently about its nominal value. Table VIII reports the results for both perturbation directions and the direction producing the largest RMSE.

Vehicle mass was the most influential parameter, producing a maximum RMSE increase of 116.2%, because it enters both the vertical-force constraint and the acceleration-induced load-transfer moments. Longitudinal CG position and CG height produced maximum increases of 24.9% and 21.5%, respectively, whereas track-width uncertainty increased RMSE by at most 7.7%. All suspension, initial-load, and aerodynamic

TABLE VII
ABLATION STUDY: AGGREGATE WHEEL-LOAD ERROR METRICS POOLED
ACROSS ALL FOUR WHEELS (NÜRBURGRING SCENARIO).

Case	RMSE [N]	RMSE [%]	P_{95} [N]	Max [N]	Δ RMSE [%]
Baseline (all ON)	233.0	5.70	460.3	2161.2	—
No Aero	237.9	5.82	496.2	2067.1	+2.1
No ARB	236.4	5.78	465.9	2112.1	+1.5
Symmetric Damper	232.9	5.70	460.0	2159.8	−0.0
No Constraints	523.0	12.79	1053.3	3324.3	+124.5

perturbations increased RMSE by no more than 3.1%. The small improvements observed for some perturbations reflect partial compensation for residual modelling errors rather than improved parameter identification. Accurate mass and CG information is therefore more important than precise suspension or aerodynamic parameter identification.

b) Sensor-noise and drift robustness: Synthetic errors were added to the accelerometer and suspension-displacement signals before causal preprocessing. Accelerometer errors comprised band-limited white noise, constant bias, and bias random walk, with nominal values representative of an automotive-grade MTi-1-class MEMS IMU. Suspension-displacement errors included linearity, repeatability, and quantization effects representative of an ILPS-19-class sensor. The same displacement-error realization was applied to the wheel-travel signals used by the anti-roll-bar model.

A common severity factor $s \in \{0, 1, 2, 4\}$ scaled all error components, where $s = 0$ denotes uncorrupted CarMaker signals and $s = 1$ denotes the nominal sensor models. For each nonzero scale, 50 independent Monte Carlo trials were performed. Metrics were pooled across all four wheels within each trial, and the median and 95th percentile were then calculated across trials.

Nominal sensor errors ($s = 1$) produced a negligible median RMSE increase of 0.0%. Even at $s = 4$, the median RMSE increased by only 0.5%, while the 95th-percentile RMSE and maximum error were 3.3% and 1.6% above their uncorrupted values, respectively. The tested sensor errors were therefore effectively attenuated by the causal preprocessing and physics-constrained update. Within the investigated ranges, the estimator was substantially more sensitive to mass and CG uncertainty than to accelerometer or suspension-displacement errors.

V. CONCLUSION

This paper presented a physics-constrained Kalman filtering framework for estimating the four-wheel vertical-load distribution using suspension and IMU measurements. The force state is not obtained from the suspension model alone: suspension-transmitted force estimates provide corner-level pseudo-measurements, while the IMU-based constraint stage uses the total vehicle mass to enforce vertical-force, pitch-moment, and roll-moment consistency across all four wheels. The resulting output is therefore an IMU-constrained wheel vertical-load estimate rather than simply the suspension-to-body force calculated independently at each corner. Aerodynamic vertical-load effects were also incorporated into the

equilibrium constraints to reduce speed-dependent bias at higher vehicle speeds.

Real-time execution on the dSPACE LabBox using CarMaker-generated ISO double-lane-change, full-lap Nürburgring, and Hockenheim signals, together with matched causal MATLAB evaluations, demonstrated consistent overall performance. In the moderate ISO double lane change, the suspension-only reconstruction and per-corner stochastic observer achieved the lowest errors because the maneuver was dominated by short-duration local suspension excitation. Under the stronger and more sustained Nürburgring excitation, however, the proposed method achieved the lowest RMSE and P_{95} errors at all four wheels and retained this advantage on Hockenheim.

The causal ablation study identified the global constraint update as the principal contributor to robustness. Removing the constraints increased the aggregate RMSE and P_{95} error by 124.5% and 128.8%, respectively. Aerodynamic compensation and anti-roll-bar coupling provided smaller additional RMSE improvements, whereas damper asymmetry had no practically significant effect under the evaluated conditions. The parameter-sensitivity study showed that vehicle mass and CG properties are the most influential model quantities. In contrast, the nominal simulated sensor-error profile increased the median RMSE by only 0.02%, and even the fourfold error scale increased the median RMSE by less than 0.5%.

Overall, the proposed method provides a computationally lightweight and extensible framework for online four-wheel vertical-load estimation. The dSPACE LabBox implementation executed the estimator at 100.0 Hz without deadline misses. Nevertheless, the real-time experiments used CarMaker-generated signals and therefore do not establish estimation accuracy under physical sensor, road, tire, suspension, and vehicle conditions. The parameter and sensor-uncertainty studies provide controlled evidence of robustness but do not replace real-vehicle validation. Future work will therefore focus on vehicle experiments using independent four-corner load measurements, implementation on a production-representative ECU, and extension to active-suspension and integrated chassis-control applications.

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TABLE VIII

MODEL-PARAMETER SENSITIVITY OVER THE NÜRBURGRING SCENARIO. RANGES CONTAIN THE RESULTS FROM BOTH PERTURBATION DIRECTIONS.

Parameter	Variation	RMSE [N]	P_{95} [N]	Maximum [N]	Worst Δ_{RMSE} [%]
Baseline	Nominal	233.0	460.3	2161.2	–
Vehicle mass	$\pm 10\%$	399.7–503.6	707.7–832.3	2093.0–2500.7	+116.2 (–10%)
CG height	$\pm 10\%$	270.2–283.0	505.2–579.6	1974.2–2348.3	+21.5 (+10%)
Longitudinal CG position	± 50.0 mm	217.8–291.1	431.5–549.5	2183.6–2306.5	+24.9 (rearward)
Suspension-force parameters	$\pm 10\%$	228.6–240.1	456.2–469.8	2151.9–2170.6	+3.1 (–10%)
Initial static wheel loads	$\pm 5\%$	230.5–239.3	457.4–467.8	2117.0–2205.5	+2.7 (–5%)
Aerodynamic coefficient	$\pm 20\%$	230.3–237.4	456.6–467.8	2140.8–2181.7	+1.9 (+20%)
Aerodynamic-centre position	± 100.0 mm	230.3–235.7	455.7–464.9	2155.4–2167.0	+1.2 (forward)
Track width	$\pm 5\%$	234.8–250.9	452.8–508.2	2067.7–2262.5	+7.7 (–5%)

TABLE IX

AGGREGATE MONTE CARLO ERRORS FOR INCREASING SENSOR-ERROR SEVERITY OVER THE NÜRBURGRING SCENARIO.

s	RMSE [N]		P_{95} [N]		Maximum [N]	
	Median	95th	Median	95th	Median	95th
0	233.0	–	460.3	–	2161.2	–
1	233.0	234.1	460.3	462.0	2161.6	2173.1
2	233.3	236.4	460.8	466.1	2163.3	2177.5
4	234.1	240.7	462.4	474.1	2163.1	2196.6

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